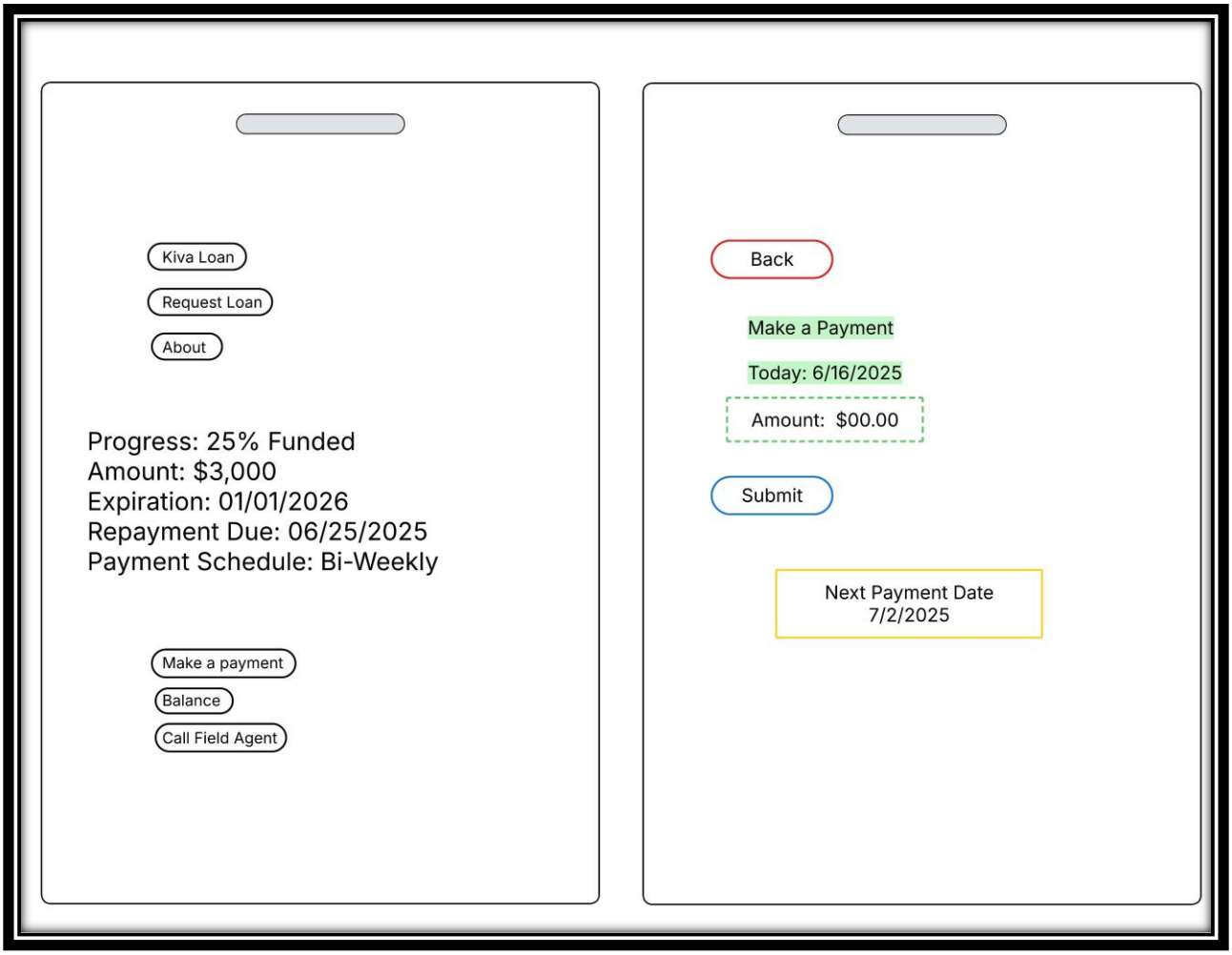
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CS 319 Project Two

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Kiva Borrower Wireframe



The interface is designed to be intuitive and accessible for users on embedded devices like flip phones or minimal web apps. It follows the principle of chunking, organizing content into digestible sections of loan progress, amount, expiration date, and repayment schedule so users can quickly grasp their financial status.

Navigation relies on clear visual signifiers. For example, the Kiva logo links to the main site, while key actions like requesting or repaying a loan are presented as buttons with strong affordances—visual cues that suggest how to interact with them. In a menu-based system, options may be numbered (e.g., 1 – Kiva, 2 – About, 3 – Request Loan), while the web version uses a wireframe with rounded buttons and accordion stacks.

The design prioritizes ease of use by limiting options and eliminating clutter. This enhances cognitive fluency and allows users to focus on essential tasks. A logical layout supports recognition over recall, a vital feature for users with limited technical literacy. For example, menu items are clearly labeled, and static information is visually separated.

Accessibility and simplicity are embedded throughout, making the interface suitable for unstructured environments like USSD menus. This minimal approach also supports seamless integration with cloud-based services and web apps, maximizing performance even on low-end hardware. As a result, Kiva and its lenders can reach underserved users cost-effectively.

Purpose and Function of Each Element and Screen

Each element in the wireframe serves a focused function tailored to device limitations. The Kiva logo acts as both a brand identifier and a navigation anchor. The About section fosters trust by outlining Kiva's mission. The Request Loan button enables borrowers to begin applications with minimal steps. Loan details are grouped into categories—progress, amount, expiration, schedule—for quick scanning and reduced cognitive load. Buttons like Make a Payment or Call a Field Agent offer direct access to core tasks, following usability principles like feedback and user control.

When users select Make a Payment, they're taken to a screen showing payment history and options. Access to this screen is available via USSD (by dialing a specific number) or, when possible, through a higher-bandwidth web app interface. Both versions share the same layout to preserve consistency and minimize system strain on resource-constrained devices like flip phones or tablets.

Benefits for the Borrower

Borrowers benefit from a streamlined interface designed for clarity on constrained hardware. Mobile-first principles like clear touch targets, minimal content per screen, and action-oriented flows to ensure that users with limited digital literacy can confidently manage loans, track progress, and request support. By emphasizing recognition over recall and reducing input demands, the design aligns with usability heuristics suitable for low-resource contexts.

Benefits for Kiva

For Kiva, this design boosts user engagement and trust in underserved regions. The minimalist interface delivers critical features without requiring modern devices, allowing broader

outreach. The self-explanatory layout reduces support overhead, while transparency via the About section strengthens brand integrity. Increased usability supports higher repayment rates and return borrowing, contributing to long-term sustainability.

Innovative Solutions Developed to Adapt to Challenges

The wireframe design for the Kiva microloan interface on the Nokia 2780 Flip applies established human interface principles from both Apple's Human Interface Guidelines and Google's Material Design to meet the constraints of low-resolution, non-touchscreen mobile devices. With a 240x320 pixel screen and limited input via a T9 keypad, the interface leverages chunking and visual hierarchy to present only the most essential information—loan amount, time remaining, and repayment interval—in clearly delineated sections. According to Apple (2024), effective interfaces reduce cognitive load by focusing attention on relevant tasks and minimizing distractions, a principle reflected in the reduced line count and screen elements. Similarly, Google's Material Design (2023) emphasizes clarity, usability, and accessibility, recommending that navigation be intuitive and that actions be clearly labeled and easily reachable, especially on constrained interfaces. By implementing a menu that uses only six to eight lines and presents options with direct numeric affordances (e.g., "1 – Request Loan"), this design enhances usability for users with limited technical literacy while aligning with best practices for recognition over recall. Rather than adapting a complex desktop interface to a small screen, the design reimagines the user's flow to match the affordances of the device, demonstrating innovative minimalism grounded in mobile-first UX research. Research from the Nielsen Norman Group provided foundational heuristics such as recognition over recall, user control, and minimalist design that were directly applied to the wireframe structure. Apple's Human Interface Guidelines and Android's Material Design principles reinforced decisions around affordance,

legibility, and button sizing for low-resolution displays. The UX Design Institute glossary helped define and apply industry-standard design language, ensuring each element had a clear cognitive and functional purpose. Studies on mobile UI in developing markets influenced the need for clear hierarchy, low-bandwidth design, and local-first functionality, all of which shaped the system's final form. The initial borrower-focused interface, designed for low-resource devices, can be effectively adapted into a cloud-based application for Kiva lenders by focusing on clarity, accessibility, and scalability. This transition retains key elements such as loan progress, repayment schedules, and borrower stories, while shifting emphasis to the needs of lenders—tracking impact, portfolio performance, and loan repayment reliability. To align with Kiva's mission of financial inclusion and transparency, the design should highlight real-time loan updates, personalized recommendations, and borrower success stories. Modular components such as a cloud-hosted loan database, analytics engine, secure payment gateway, and notification system must communicate via APIs using event-driven, asynchronous architecture to ensure responsive, real-time updates.

Different lender types have varied goals: philanthropists want personal borrower updates, impact investors seek performance metrics, and casual donors need simplicity. The design addresses these needs through customizable dashboards and intuitive navigation.

Next steps include mapping lender user journeys, creating wireframes, defining APIs, building a secure cloud backend, and conducting usability testing. By extending core design principles of chunking, minimalism, and task-focused flow, the system can support a scalable, cloud-based platform that meets the needs of both borrowers and lenders while staying true to Kiva's mission.

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